

# COLLINS CULBERT

## Electrical Systems Engineering Student · Embedded Systems & Robotics Control

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### SUMMARY

Junior Electrical Systems Engineering Technology student with hands-on experience across closed-loop robotics control, FPGA digital logic, and register-level embedded programming. Led the engineering build of a **10-servo, PID-controlled robotic hand** that earned a **1st Place — Best Presentation award** and finalist standing. Comfortable working end to end: schematic to firmware to live demo.

### TECHNICAL SKILLS

|                             |                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>          | Python, SQL, C/C++ (STM32 bare metal), MicroPython, JavaScript, HTML/CSS                                                       |
| <b>Hardware &amp; Tools</b> | STM32F4xx / Keil uVision, Quartus Prime (FPGA/DE10), KiCad, oscilloscopes, multimeters, DC motor drivers                       |
| <b>Specializations</b>      | Closed-loop PID control, I <sup>2</sup> C sensor integration, FPGA state machine design, register-level GPIO/timer programming |
| <b>Software</b>             | MS Office Suite, POS systems, HubSpot (basics)                                                                                 |

### RELEVANT COURSEWORK

**ESET 210** – Circuit Analysis · **ESET 269** – Embedded Systems Development in C · **ESET 369** – Embedded Systems Software · **PHYS 207** – Electricity & Magnetism · **MATH 251** – Calculus III

### PROJECTS

#### Robotic Hand — Closed-Loop PID Control

*MicroPython, RP2040/ESP32, AS5600, PCA9548A, I<sup>2</sup>C*

- ▶ Co-designed and built a **10-servo DC motor-driven robotic hand** (2 continuous-rotation servos per finger) for the Texas A&M Robotic Hand Challenge.
- ▶ Engineered a **closed-loop PID control architecture** using **AS5600 magnetic encoders** read over I<sup>2</sup>C through a PCA9548A multiplexer for per-finger position feedback.
- ▶ Built a **calibration workflow** in MicroPython to capture and store named hand poses from live encoder data.
- ▶ Delivered the technical presentation that earned the team the **1st Place — Best Presentation Award** and overall finalist standing.

#### FPGA Motor Speed Controller

*Quartus Prime, DE10 FPGA, Verilog/VHDL, PWM*

- ▶ Designed and implemented a **7-state, one-hot Moore state machine** PWM motor speed controller on the DE10 FPGA.
- ▶ Drove a **BCD 7-segment display** output directly from FSM state logic.

#### STM32F4xx Embedded Systems Lab

*C (bare metal), STM32F4xx, Keil uVision*

- ▶ Programmed **GPIO peripherals and SysTick timers** at the register level in C with no HAL abstraction.

### EDUCATION

#### B.S., Electrical Systems Engineering Technology (in progress)

Texas A&M University, College Station, TX · **Regents Scholar**

*Expected Dec. 2028*

### LEADERSHIP & INVOLVEMENT

#### Cohort 2 Member, KCLI Brotherhood

*College Station, TX · Sept. 2024 – Present*

- ▶ Oversee compliance with organization and chapter safety rules and regulations for **20+ members**.